3.2.4 Properties of Period 3 elements and their oxides (A-level only)

The reactions of the Period 3 elements with oxygen are considered. The pH of the solutions formed when the oxides react with water illustrates further trends in properties across this period. Explanations of these reactions offer opportunities to develop an in-depth understanding of how and why these reactions occur.

<table>
<tr><th>Content</th><th>Opportunities for skills development</th></tr>
<tr><td>
The reactions of Na and Mg with water.
The trends in the reactions of the elements Na, Mg, Al, Si, P and S with oxygen, limited to the formation of Na_2O, MgO, Al_2O_3, SiO_2, P_4O_{10}, SO_2 and SO_3
The trend in the melting point of the highest oxides of the elements Na–S
The reactions of the oxides of the elements Na–S with water, limited to Na_2O, MgO, Al_2O_3, SiO_2, P_4O_{10}, SO_2 and SO_3, and the pH of the solutions formed.
The structures of the acids and the anions formed when P_4O_{10}, SO_2 and SO_3 react with water.
Students should be able to:

explain the trend in the melting point of the oxides of the elements Na–S in terms of their structure and bonding
explain the trends in the reactions of the oxides with water in terms of the type of bonding present in each oxide
write equations for the reactions that occur between the oxides of the elements Na–S and given acids and bases.

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AT a, c and k
PS 2.2
Students could carry out reactions of elements with oxygen and test the pH of the resulting oxides.
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